

Curve Fitting

Curve fitting is a fundamental technique in statistics and data analysis used to **construct a mathematical function** that best represents the relationship between a set of observed data points. The goal is to find a curve that passes as close as possible to all the given data points.



WTF Is Curve Fitting In First Place

Imagine you have collected some **data points**, maybe the price of a stock over several days, or the temperature of a cooling object over time. If you plot these points on a graph, they might show a certain **pattern** or **trend**.

Curve fitting is the process of finding a **formula** (an equation, like $y = 2x + 5$ or $y = 3x^2$) whose graph (the "curve") closely matches that pattern in your data. It's like drawing the best smooth line or curve through a scatter of dots.

- **Input:** A set of **paired data points** $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$.
- **Process:** Choosing a specific type of function (e.g., a straight line, a parabola, an exponential curve) and finding the specific **parameters** (like the slope and intercept of a line) that make the function's graph fit the data best.
- **Output:** The **equation** of the "best-fit" curve.



Why U Fit Curves

There are two main reasons for fitting curves to data:

1. **Estimation and Prediction:** Once you have the equation of the curve, you can use it to **estimate** the value of y for an **intermediate** value of x (interpolation) or **predict** the value of y for a value of x **outside** the range of your original data (extrapolation). For example, if your curve fits the relationship between years of schooling (x) and income (y), you can predict the income of someone with a certain number of years of schooling.
2. **Summarizing and Understanding the Relationship:** The equation itself provides a concise mathematical **model** of the relationship between the variables. The parameters of the fitted curve (like the slope) give insight into the nature and strength of that relationship.



How They're Fitted

Curves are fitted by defining a measure of how "good" the fit is and then trying to **minimize** the badness of the fit.

- **The Problem:** When you draw a curve through data points, most of the points will **not** lie exactly on the curve. The vertical distance between an actual data point (y_{actual}) and the corresponding point on the fitted curve ($\hat{y}_{predicted}$) is called the **residual** or **error** (e). $e = y_{actual} - \hat{y}_{predicted}$.
- **The Goal:** We want to find the curve whose equation makes the total size of all these errors (residuals) as **small** as possible.
- **The Common Technique:** The most widely used technique is the **Method of Least Squares**.



Method Of Least Squares

The **Method of Least Squares** is the standard procedure for finding the curve that best fits a set of data.

- **The Idea:** Simply adding up the errors is not useful because positive errors (points above the curve) and negative errors (points below the curve) would cancel each other out.
- **The Solution:** To overcome the cancellation problem, we **square** each error (residual) before summing them up. Squaring the errors ensures that:
 1. All contributions to the total "badness of fit" are **positive**.
 2. Larger errors are **magnified** (a residual of 10 squared is 100, a residual of 1 squared is 1). This heavily penalizes curves that miss a few points by a large distance.
- **The Formula:** The quantity we seek to minimize is the **Sum of Squared Errors (SSE)** or S :

$$S = \sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

where y_i is the actual y -value and $\hat{y}_i$ is the y -value predicted by the fitted curve.

- **The Technique:** We treat the SSE (S) as a function of the unknown parameters (e.g., a and b for a straight line $y = a + bx$). To find the values of the parameters that **minimize** S , we use calculus:
 1. We take the **partial derivative** of S with respect to each unknown parameter.
 2. We set each of these derivatives equal to **zero**.
 3. This results in a system of equations called the **Normal Equations**.
 4. We solve the Normal Equations simultaneously to find the optimal values for the parameters.

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Fitting A Straight Line

The simplest and most common type of curve to fit is a **straight line**, also known as **Linear Regression**. The equation for a straight line is:

$$\hat{y} = a + bx$$

Where:

- $\hat{y}$ is the **predicted** value of the dependent variable.
- x is the value of the independent variable.
- a is the **y-intercept** (the predicted value of $\hat{y}$ when $x = 0$).
- b is the **slope** (how much $\hat{y}$ changes for a one-unit change in x).

Our goal is to find the values of a and b that minimize the Sum of Squared Errors, $S = \sum (y_i - \hat{y}_i)^2 = \sum (y_i - (a + bx_i))^2$.

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Normal Equations For A Straight Line

To find the values of a and b that minimize S , we take the partial derivatives of S with respect to a and b and set them to zero. This process yields the following two simultaneous equations, called the **Normal Equations for a Straight Line**:

1.
$$\sum y = na + b \sum x$$

2.
$$\sum xy = a \sum x + b \sum x^2$$

Where:

- n is the number of data points.
- $\sum y$ is the sum of all the y -values.
- $\sum x$ is the sum of all the x -values.
- $\sum xy$ is the sum of the products of each x_i and y_i .
- $\sum x^2$ is the sum of the squares of each x_i .

By calculating the four sums ($\sum x$, $\sum y$, $\sum x^2$, $\sum xy$) from the raw data and substituting them into these two equations, we can solve for the two unknowns, a and b .



Trick To Remember

The Normal Equations can be conceptually remembered by starting with the original equation of the line, $\hat{y} = a + bx$, and performing two operations:

1. **First Equation:** Sum the entire equation across all n data points.

$$\sum \hat{y} = \sum (a + bx)$$

Since the sum of the predicted $\hat{y}$ values is approximately the same as the sum of the actual y values ($\sum \hat{y} \approx \sum y$), and since a is a constant summed n times, we get:

$$\sum y = na + b \sum x$$

2. **Second Equation:** Multiply the original equation by x and then sum the resulting terms across all n data points.

$$x\hat{y} = x(a + bx) \implies x\hat{y} = ax + bx^2$$

Summing this (and replacing $\sum x\hat{y}$ with $\sum xy$):

$$\sum xy = a \sum x + b \sum x^2$$



How To Create Normal Equation Of Any Straight Line

The general method for deriving the normal equations for **any** model (not just a straight line) is based on the parameters of the model itself.

For the model $\hat{y} = a + b \cdot f(x)$, where $f(x)$ is a function of x :

1. **Identify the parameters** to be estimated. For $\hat{y} = a + bx$, the parameters are a and b .
2. **Identify the "coefficients"** of the parameters in the equation.
 - The coefficient of a is 1 .
 - The coefficient of b is x .
3. **The Rule:** The Normal Equations are created by:
 - **Equation 1 (for parameter a):** Multiply the original equation by the coefficient of a (which is 1) and then sum.

$$\sum(\hat{y} \cdot 1) = \sum((a + bx) \cdot 1) \implies \sum y = na + b \sum x$$

- **Equation 2 (for parameter b):** Multiply the original equation by the coefficient of b (which is x) and then sum.

$$\sum(\hat{y} \cdot x) = \sum((a + bx) \cdot x) \implies \sum xy = a \sum x + b \sum x^2$$

This rule is a powerful shortcut for writing the Normal Equations for other types of curves as well.



Solve Multiple Different Cases Of Normal Equation Writing for Straight lines

Let's apply the rule to slight variations of the straight-line model.

Case 1: Standard Straight Line

- **Model:** $\hat{y} = a + bx$
- **Parameters:** a (coefficient 1), b (coefficient x).
- **Normal Equations:**
 1. Multiply by 1 : $\sum y = na + b \sum x$
 2. Multiply by x : $\sum xy = a \sum x + b \sum x^2$

Case 2: Line Through the Origin

- **Model:** $\hat{y} = bx$ (The y -intercept a is forced to be 0).
- **Parameter:** b (coefficient x).
- **Normal Equation (Only one equation is needed to solve for one unknown b):**
 1. Multiply by the coefficient of b , which is x :

$$\begin{aligned}\sum(\hat{y} \cdot x) &= \sum(bx \cdot x) \\ \sum xy &= b \sum x^2\end{aligned}$$

Case 3: A Simple Shifted Curve (Example for the power of the method)

- **Model:** $\hat{y} = a + b(\log x)$ (This is a straight line if plotted against $\log x$).
- **Parameters:** a (coefficient 1), b (coefficient $\log x$).
- **Normal Equations:**
 1. Multiply by 1:

$$\sum y = na + b \sum \log x$$

2. Multiply by $\log x$:

$$\sum(y \log x) = a \sum \log x + b \sum (\log x)^2$$

This shows the method works for any linear combination of parameters, where you just substitute the coefficient term.

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That's a great way to practice setting up the equations! The key, as discussed, is identifying the parameters and their coefficients.

Here are two examples for linearizable models and two examples for models that require the simple Normal Equations structure, focusing only on the creation of the Normal Equations.

1. Linearizable Models (Linear in Parameters)

These models are linear if we define new variables based on the original x and y . We will treat the coefficients of the parameters as the terms used to create the Normal Equations.

Example 1.1: Inverse and Power Fit

- **Model:** $\hat{y} = A + B \cdot \frac{1}{x^2}$
 - This is a straight line if plotted against $\frac{1}{x^2}$.
 - **Parameters to solve for:** A and B .
 - **Coefficient of A :** 1.
 - **Coefficient of B :** $\frac{1}{x^2}$.

Normal Equations:

1. **Multiply by 1 (coefficient of A):**

$$\sum y = nA + B \sum \frac{1}{x^2}$$

2. **Multiply by $\frac{1}{x^2}$ (coefficient of B):**

$$\sum \left(y \cdot \frac{1}{x^2} \right) = A \sum \frac{1}{x^2} + B \sum \left(\frac{1}{x^2} \cdot \frac{1}{x^2} \right)$$

$$\sum \frac{y}{x^2} = A \sum \frac{1}{x^2} + B \sum \frac{1}{x^4}$$

Example 1.2: Exponential Growth/Decay (After Transformation)

- **Original Model (Non-Linear):** $\hat{y} = ae^{bx}$
- **Linearization Step:** Take the natural logarithm ($\ln$) of both sides.

$$\ln(\hat{y}) = \ln(a) + bx$$

- **Linear Form:** Let $Y = \ln(\hat{y})$, and let $A = \ln(a)$.

$$Y = A + bx$$

- **Parameters to solve for:** A and b .
- **Coefficient of A :** 1.
- **Coefficient of b :** x .

Normal Equations:

1. **Multiply by 1 (coefficient of A):**

$$\sum Y = nA + b \sum x$$

2. **Multiply by x (coefficient of b):**

$$\sum(Yx) = A \sum x + b \sum x^2$$

(Note: Once A and b are solved, the original parameter a is found by $a = e^A$.)

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2. Polynomial/Multiple Regression Models (Still Linear in Parameters)

These models are also formally **linear in the parameters**, a, b, c , etc., and require the standard least squares technique, resulting in as many Normal Equations as there are parameters.

Example 2.1: Quadratic (Parabolic) Fit

- **Model:** $\hat{y} = a + bx + cx^2$
 - **Parameters to solve for:** a , b , and c .
 - **Coefficient of a :** 1.
 - **Coefficient of b :** x .
 - **Coefficient of c :** x^2 .

Normal Equations (3 equations for 3 unknowns):

1. **Multiply by 1 (coefficient of a):**

$$\sum y = na + b \sum x + c \sum x^2$$

2. **Multiply by x (coefficient of b):**

$$\sum (y \cdot x) = a \sum x + b \sum (x \cdot x) + c \sum (x^2 \cdot x)$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

3. **Multiply by x^2 (coefficient of c):**

$$\sum (y \cdot x^2) = a \sum x^2 + b \sum (x \cdot x^2) + c \sum (x^2 \cdot x^2)$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

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Example 2.2: Line with a Logarithmic Term

- **Model:** $\hat{y} = C_1 + C_2 \log(x)$
 - This is a straight line if plotted against $\log(x)$.
 - **Parameters to solve for:** C_1 and C_2 .
 - **Coefficient of C_1 :** 1.
 - **Coefficient of C_2 :** $\log(x)$.

Normal Equations:

1. **Multiply by 1 (coefficient of C_1):**

$$\sum y = nC_1 + C_2 \sum \log(x)$$

2. **Multiply by $\log(x)$ (coefficient of C_2):**

$$\sum (y \cdot \log x) = C_1 \sum \log x + C_2 \sum (\log x \cdot \log x)$$

$$\sum y \log x = C_1 \sum \log x + C_2 \sum (\log x)^2$$

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Fitting Of Exponential Curves

Fitting of Exponential Curves

An exponential curve describes a relationship where a change in the independent variable (x) leads to a proportional change in the dependent variable (y). The most common forms are $\hat{y} = ab^x$ and $\hat{y} = ae^{bx}$.

A. General Equation Creation (Model 1: $\hat{y} = ab^x$)

The goal is to transform the non-linear model $\hat{y} = ab^x$ into a straight-line form, $Y = A + Bx$, so we can use the Normal Equations.

1. Linearization Step

Take the logarithm (usually the natural log, $\ln$, or log base 10, $\log$) of both sides of the original equation:

$$\hat{y} = ab^x$$

Using the **Product Rule** and the **Power Rule** of logarithms:

$$\log(\hat{y}) = \log(ab^x)$$

$$\log(\hat{y}) = \log(a) + \log(b^x)$$

$$\log(\hat{y}) = \log(\mathbf{a}) + \mathbf{x} \log(\mathbf{b})$$

2. Defining New Variables

We define new variables to match the straight-line equation $Y = A + Bx$:

- **New Dependent Variable (Y):** $Y = \log(\hat{y})$
- **New Y-Intercept (A):** $A = \log(a)$
- **New Slope (B):** $B = \log(b)$
- **New Independent Variable (X):** $X = x$

The linear form is: $\mathbf{Y} = \mathbf{A} + \mathbf{B}\mathbf{X}$

3. Normal Equations for the Linearized Form

Since the new equation $Y = A + BX$ is a straight line, we use the standard Normal Equations (with Y replacing y and X replacing x):

$$1. \quad \sum Y = nA + B \sum X$$

$$2. \quad \sum XY = A \sum X + B \sum X^2$$

4. Finding the Original Parameters (a and b)

After solving the Normal Equations to find the values for A and B , you must revert them back to the original parameters, a and b , using the inverse operation of the logarithm you used.

- If you used **ln**: $\mathbf{a} = e^A$ and $\mathbf{b} = e^B$

- If you used $\log_{10}$: $a = 10^A$ and $b = 10^B$

or

B. General Equation Creation (Model 2: $\hat{y} = ae^{bx}$)

This model is common when using natural growth/decay rates.

1. Linearization Step

Take the **Natural Log (ln)** of both sides of the original equation:

$$\hat{y} = ae^{bx}$$

$$\ln(\hat{y}) = \ln(ae^{bx})$$

$$\ln(\hat{y}) = \ln(a) + \ln(e^{bx}) \quad (\text{Product Rule})$$

$$\ln(\hat{y}) = \ln(a) + bx \quad (\text{Since } \ln(e^k) = k)$$

2. Defining New Variables

- **New Dependent Variable (Y):** $Y = \ln(\hat{y})$
- **New Y-Intercept (A):** $A = \ln(a)$
- **New Slope (B):** $B = b$
- **New Independent Variable (X):** $X = x$

The linear form is: $Y = A + BX$

3. Normal Equations

Since the linear form is identical to Model 1, the Normal Equations are the same:

$$1. \quad \sum Y = nA + B \sum X$$

2.

$$\sum XY = A \sum X + B \sum X^2$$

4. Finding the Original Parameters (a and b)

The parameter B is the same as b , and A is converted back to a :

- $b = B$
- $a = e^A$

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Example: Fitting $\hat{y} = ab^x$

Let's find the Normal Equations and the required sums for the data below to fit the curve $\hat{y} = ab^x$. (We will use $\log_{10}$ for this example.)

x	y
1	3.0
2	4.8
3	7.6
4	12.1

Step 1: Linearization and Identifying Variables

- **Linear Form:** $\log_{10}(\hat{y}) = \log_{10}(a) + x \log_{10}(b)$
- **New Variables:**
 - $Y = \log_{10}(y)$
 - $A = \log_{10}(a)$
 - $B = \log_{10}(b)$
 - $X = x$

Step 2: Calculate the Required Sums

We need sums of X , Y , X^2 , and XY . Since $X = x$, we need $\sum x$, $\sum \log_{10}(y)$, $\sum x^2$, and $\sum x \log_{10}(y)$.

x (X)	y	$\log_{10}(y)$ (Y)	x^2 (X^2)	$x \cdot \log_{10}(y)$ (XY)	
1	3.0	0.4771	1	0.4771	
2	4.8	0.6812	4	1.3624	
3	7.6	0.8808	9	2.6424	
4	12.1	1.0828	16	4.3312	
$n = 4$		$\sum Y = 3.1219$	$\sum X = 10$	$\sum X^2 = 30$	$\sum XY = 8.8131$

Step 3: Write the Normal Equations

Substitute the sums into the Normal Equations (using A and B):

1.
$$\sum Y = nA + B \sum X$$
$$3.1219 = 4A + 10B \quad (\text{Equation 1})$$

2.
$$\sum XY = A \sum X + B \sum X^2$$
$$8.8131 = 10A + 30B \quad (\text{Equation 2})$$

Step 4: Solve for A and B

1. Multiply Equation 1 by 2.5 to match the A coefficient in Equation 2:

$$2.5 \times (3.1219) = 2.5 \times (4A) + 2.5 \times (10B)$$
$$7.80475 = 10A + 25B \quad (\text{Equation 3})$$

2. Subtract Equation 3 from Equation 2:

$$(8.8131 - 7.80475) = (10A - 10A) + (30B - 25B)$$

$$1.00835 = 5B$$

$$B = 0.20167$$

3. Substitute B back into Equation 1:

$$3.1219 = 4A + 10(0.20167)$$

$$3.1219 = 4A + 2.0167$$

$$4A = 3.1219 - 2.0167 = 1.1052$$

$$A = 0.2763$$

Step 5: Find Original Parameters (a and b)

Since we used $\log_{10}$:

- $a = 10^A = 10^{0.2763} \approx 1.89$

- $b = 10^B = 10^{0.20167} \approx 1.59$

Final Exponential Equation

The best-fit exponential curve is:

$$\hat{y} = 1.89 \cdot (1.59)^x$$

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That's a great next step! Fitting a **Second Order Parabola** (also known as a **Quadratic Curve**) is a perfect example of fitting a curve that is **linear in its parameters** but goes beyond a simple straight line.

Fitting a Second Order Parabola

A second order parabola is represented by a polynomial of degree two. It's used when the relationship between x and y is **curvilinear** (U-shaped or inverted U-shaped).

1. The Model Equation

The general equation for a parabola is:

$$\hat{y} = a + bx + cx^2$$

- **Parameters to solve for:** a (Y-intercept), b (coefficient of the linear term), and c (coefficient of the quadratic term).
- Since the parameters a , b , and c all appear linearly (they are not exponents or multiplied by each other), we can use the Method of Least Squares directly, resulting in **three** Normal Equations.

[Image of a scatter plot with a parabolic curve drawn through the data points representing quadratic curve fitting]

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2. General Equation Creation (Normal Equations)

We have three unknown parameters (a , b , and c), so we need three simultaneous Normal Equations. We derive these by multiplying the original model equation by the coefficient of each parameter and then summing the result.

Parameter	Coefficient in $\hat{y} = a + bx + cx^2$	Operation (Multiply and Sum)
a	1	$\sum(\hat{y} \cdot 1) = \sum(a \cdot 1 + bx \cdot 1 + cx^2 \cdot 1)$
b	x	$\sum(\hat{y} \cdot x) = \sum(a \cdot x + bx \cdot x + cx^2 \cdot x)$
c	x^2	$\sum(\hat{y} \cdot x^2) = \sum(a \cdot x^2 + bx \cdot x^2 + cx^2 \cdot x^2)$

The Three Normal Equations

Setting $\sum \hat{y} \approx \sum y$, the resulting system is:

1.
$$\sum y = na + b \sum x + c \sum x^2$$

$$2. \quad \sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$3. \quad \sum x^2y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

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3. Required Calculations

To solve this system, you must compute the following seven sums from your raw data points (x_i, y_i) :

$$n, \quad \sum x, \quad \sum y, \quad \sum x^2, \quad \sum x^3, \quad \sum x^4, \quad \sum xy, \quad \sum x^2y$$

4. Solving the System

Once all the required sums are calculated and substituted into the three Normal Equations, you must solve the resulting 3×3 system of linear equations simultaneously for the unknowns a , b , and c . This can be done using techniques like substitution, elimination, or matrix methods.

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Example: Creation of Normal Equations for Parabola

Let's use a small hypothetical dataset to set up the Normal Equations for $\hat{y} = a + bx + cx^2$.

x	y	x^2	x^3	x^4	xy	x^2y
0	1.2	0	0	0	0.0	0.0
1	0.8	1	1	1	0.8	0.8

x	y	x^2	x^3	x^4	xy	x^2y
2	0.3	4	8	16	0.6	1.2
3	0.5	9	27	81	1.5	4.5
Total	$\sum y = 2.8$	$\sum x = 6$	$\sum x^2 = 14$	$\sum x^3 = 36$	$\sum x^4 = 98$	$\sum xy = 2.9$

- $n = 4$

The Normal Equations with Substituted Values:

$$1. \quad \sum y = na + b \sum x + c \sum x^2$$

$$2.8 = 4a + 6b + 14c$$

$$2. \quad \sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$2.9 = 6a + 14b + 36c$$

$$3. \quad \sum x^2y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

$$6.5 = 14a + 36b + 98c$$

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This model is fundamentally non-linear, but it can be easily **linearized** using logarithms, similar to the $y = ax^b$ model we solved earlier.

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Fitting the Power-Law Curve: $PV^r = C$

1. Linearization

The goal is to transform the equation into the linear form $\mathbf{Y} = \mathbf{A} + \mathbf{B}\mathbf{X}$.

Starting with the model:

$$PV^r = C$$

Take the natural logarithm ($\ln$) of both sides:

$$\ln(PV^r) = \ln(C)$$

Using the **Product Rule** and **Power Rule** of logarithms:

$$\ln(P) + \ln(V^r) = \ln(C)$$

$$\ln(\mathbf{P}) + r \ln(\mathbf{V}) = \ln(\mathbf{C})$$

Now, we rearrange this to match the straight-line form $\mathbf{Y} = \mathbf{A} + \mathbf{B}\mathbf{X}$:

$$\ln(P) = \ln(C) - r \ln(V)$$

2. Defining New Variables

- **New Dependent Variable (Y):** $Y = \ln(P)$
- **New Independent Variable (X):** $X = \ln(V)$
- **New Y-Intercept (A):** $A = \ln(C)$
- **New Slope (B):** $B = -r$ (Note the negative sign!)

The linear form is: $\mathbf{Y} = \mathbf{A} + \mathbf{B}\mathbf{X}$

3. Normal Equations

Since the model is now linear in A and B , we use the standard Normal Equations:

1.
$$\sum \mathbf{Y} = n\mathbf{A} + \mathbf{B} \sum \mathbf{X}$$

2.
$$\sum \mathbf{XY} = \mathbf{A} \sum \mathbf{X} + \mathbf{B} \sum \mathbf{X}^2$$

4. Finding Original Parameters

After solving for A and B :

- The exponent r is found by: $r = -B$
- The constant C is found by: $C = e^A$



Example 1: Finding r and C (Basic Case)

Q1. Fit the curve $PV^r = C$ to the following data:

P	100	50	20
V	1.0	2.0	4.0

1. Calculate the Required Sums

$n = 3$. We need $X = \ln(V)$, $Y = \ln(P)$, X^2 , and XY .

P	V	$X = \ln(V)$	$Y = \ln(P)$	X^2	XY
100	1.0	0.0000	4.6052	0.0000	0.0000
50	2.0	0.6931	3.9120	0.4804	2.7153
20	4.0	1.3863	2.9957	1.9218	4.1537
Total		$\sum X = 2.0794$	$\sum Y = 11.5129$	$\sum X^2 = 2.4022$	$\sum XY = 6.8690$

2. Set Up and Solve Normal Equations

1. $11.5129 = 3A + 2.0794B$ (Eq. 1)

2. $6.8690 = 2.0794A + 2.4022B$ (Eq. 2)

- From (Eq. 1): $A = \frac{11.5129 - 2.0794B}{3} = 3.8376 - 0.6931B$
- Substitute A into (Eq. 2):

$$6.8690 = 2.0794(3.8376 - 0.6931B) + 2.4022B$$

$$6.8690 = 7.9796 - 1.4404B + 2.4022B$$

$$6.8690 - 7.9796 = 0.9618B$$

$$-1.1106 = 0.9618B \implies \mathbf{B \approx -1.1547}$$

- Find A :

$$A = 3.8376 - 0.6931(-1.1547)$$

$$A = 3.8376 + 0.8000 \implies \mathbf{A \approx 4.6376}$$

3. Find Original Parameters (r and C)

- $r = -B = -(-1.1547) \implies \mathbf{r \approx 1.1547}$
- $C = e^A = e^{4.6376} \implies \mathbf{C \approx 103.55}$

Final Result

The best-fit power-law equation is:

$$\mathbf{PV^{1.1547} = 103.55}$$

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Example 2: Finding r and C (Larger Dataset)

Q2. Find the constants r and C for the equation $\mathbf{PV^r = C}$ using the following data:

P	210	165	130	100	80
V	2.0	2.5	3.0	3.5	4.0

1. Calculate the Required Sums

$n = 5$. We need $X = \ln(V)$, $Y = \ln(P)$, X^2 , and XY .

P	V	$X = \ln(V)$	$Y = \ln(P)$	X^2	XY	
210	2.0	0.6931	5.3471	0.4804	3.7077	
165	2.5	0.9163	5.1059	0.8396	4.6784	
130	3.0	1.0986	4.8676	1.2069	5.3486	
100	3.5	1.2528	4.6052	1.5695	5.7699	
80	4.0	1.3863	4.3820	1.9218	6.0754	
Total		$\sum X = 5.3471$	$\sum Y = 24.3078$	$\sum X^2 = 6.0182$	$\sum XY = 25.5800$	

2. Set Up and Solve Normal Equations

$$1. \sum Y = nA + B \sum X \implies \mathbf{24.3078} = 5A + 5.3471B \quad (\text{Eq. 1})$$

$$2. \sum XY = A \sum X + B \sum X^2 \implies \mathbf{25.5800} = 5.3471A + 6.0182B \quad (\text{Eq. 2})$$

To eliminate A , multiply (Eq. 1) by $\frac{5.3471}{5} \approx 1.06942$:

$$1.06942 \times (24.3078) = 1.06942 \times (5A) + 1.06942 \times (5.3471B)$$

$$\mathbf{25.9944 \approx 5.3471A + 5.7180B} \quad (\text{Eq. 3})$$

Subtract (Eq. 3) from (Eq. 2):

$$(25.5800 - 25.9944) = (5.3471A - 5.3471A) + (6.0182B - 5.7180B)$$

$$\mathbf{-0.4144 = 0.3002B}$$

$$\mathbf{B \approx -1.3804}$$

Find A using (Eq. 1):

$$24.3078 = 5A + 5.3471(-1.3804)$$

$$24.3078 = 5A - 7.3811$$

$$5A = 24.3078 + 7.3811 = 31.6889$$

$$\mathbf{A \approx 6.3378}$$

3. Find Original Parameters (r and C)

- $r = -B = -(-1.3804) \implies \mathbf{r \approx 1.3804}$
- $C = e^A = e^{6.3378} \implies \mathbf{C \approx 565.66}$

Final Result

The best-fit power-law equation is:

$$\mathbf{P V^{1.3804} = 565.66}$$

∞

Curve Fitting Solutions

Fitting a Straight Line (Example)

Q1. Fit a straight line to the following data by least square method.

$$\text{Model: } \hat{y} = a + bx$$

X	0	1	2	3	4
y	1	1.8	3.3	4.5	6.3

1. Calculate the Required Sums

The Normal Equations are:

1. $\sum y = na + b \sum x$
2. $\sum xy = a \sum x + b \sum x^2$

x	y	x^2	xy	
0	1.0	0	0.0	
1	1.8	1	1.8	
2	3.3	4	6.6	
3	4.5	9	13.5	
4	6.3	16	25.2	
Total	$\sum x = 10$	$\sum y = 16.9$	$\sum x^2 = 30$	$\sum xy = 47.1$

Here, $n = 5$.

2. Set Up and Solve Normal Equations

Substitute the sums into the Normal Equations:

$$1. \quad 16.9 = 5a + 10b \quad (\text{Eq. 1})$$

$$2. \quad 47.1 = 10a + 30b \quad (\text{Eq. 2})$$

To solve by elimination, multiply (Eq. 1) by 2:

$$2 \times (16.9) = 2 \times (5a) + 2 \times (10b)$$

$$33.8 = 10a + 20b \quad (\text{Eq. 3})$$

Subtract (Eq. 3) from (Eq. 2):

$$(47.1 - 33.8) = (10a - 10a) + (30b - 20b)$$

$$13.3 = 10b$$

$$\mathbf{b = 1.33}$$

Substitute $b = 1.33$ back into (Eq. 1):

$$16.9 = 5a + 10(1.33)$$

$$16.9 = 5a + 13.3$$

$$5a = 16.9 - 13.3 = 3.6$$

$$\mathbf{a = 0.72}$$

Final Result

The best-fit straight line is:

$$\hat{y} = 0.72 + 1.33x$$

or

Fitting an Exponential Curve ($y = ax^b$) (Example)

Q1. Fit the following Equation, $y = ax^b$, to the following data:

$$\text{Model: } \hat{y} = ax^b$$

X	1	2	3	4	5
y	0.5	1.7	3.4	5.7	8.4

1. Linearization

Take the natural logarithm ($\ln$) of both sides:

$$\ln(\hat{y}) = \ln(ax^b)$$

$$\ln(\hat{y}) = \ln(a) + b \ln(x)$$

Define the new variables and parameters for the linear form $Y = A + BX$:

$$Y = \ln(y), \quad A = \ln(a), \quad B = b, \quad X = \ln(x)$$

2. Calculate the Required Sums

We need sums of X , Y , X^2 , and XY . $n = 5$.

x	y	$X = \ln(x)$	$Y = \ln(y)$	$X^2 = (\ln x)^2$	$XY = (\ln x) \cdot (\ln y)$
1	0.5	0.0000	-0.6931	0.0000	0.0000
2	1.7	0.6931	0.5306	0.4804	0.3677

x	y	$X = \ln(x)$	$Y = \ln(y)$	$X^2 = (\ln x)^2$	$XY = (\ln x) \cdot (\ln y)$
3	3.4	1.0986	1.2238	1.2070	1.3444
4	5.7	1.3863	1.7405	1.9218	2.4129
5	8.4	1.6094	2.1282	2.5902	3.4241
Total		$\sum X = 4.7874$	$\sum Y = 4.9300$	$\sum X^2 = 6.1994$	$\sum XY = 7.5491$

3. Set Up and Solve Normal Equations

$$1. \sum Y = nA + B \sum X \implies 4.9300 = 5A + 4.7874B \quad (\text{Eq. 1})$$

$$2. \sum XY = A \sum X + B \sum X^2 \implies 7.5491 = 4.7874A + 6.1994B \quad (\text{Eq. 2})$$

Solving the system (using substitution/elimination or calculator/software for precision):

$$\text{From (Eq. 1): } A = \frac{4.9300 - 4.7874B}{5} = 0.9860 - 0.9575B$$

Substitute A into (Eq. 2):

$$7.5491 = 4.7874(0.9860 - 0.9575B) + 6.1994B$$

$$7.5491 = 4.7180 - 4.5841B + 6.1994B$$

$$7.5491 - 4.7180 = 1.6153B$$

$$2.8311 = 1.6153B \implies B \approx 1.7527$$

Substitute B back into the expression for A :

$$A = 0.9860 - 0.9575(1.7527) = 0.9860 - 1.6781$$

$$A \approx -0.6921$$

4. Find Original Parameters (a and b)

- $b = B \approx 1.75$
- $a = e^A = e^{-0.6921} \approx 0.5005$

Final Result

The best-fit exponential curve is:

$$\hat{y} = 0.5005x^{1.75}$$

or

Daily Quiz (Curve Fitting)

Q1. Fit a second degree parabola to the following data-

$$\text{Model: } \hat{y} = a + bx + cx^2$$

X	0	1	2	3	4
y	1	0	3	10	21

1. Calculate the Required Sums

The Normal Equations are:

- $\sum y = na + b \sum x + c \sum x^2$
- $\sum xy = a \sum x + b \sum x^2 + c \sum x^3$
- $\sum x^2y = a \sum x^2 + b \sum x^3 + c \sum x^4$

x	y	x^2	x^3	x^4	xy	x^2y	
0	1	0	0	0	0	0	
1	0	1	1	1	0	0	
2	3	4	8	16	6	12	
3	10	9	27	81	30	90	
4	21	16	64	256	84	336	
Total	$\sum y = 35$	$\sum x = 10$	$\sum x^2 = 30$	$\sum x^3 = 100$	$\sum x^4 = 354$	$\sum xy = 120$	$\sum x^2y = 438$

Here, $n = 5$.

2. Set Up Normal Equations

1. $35 = 5a + 10b + 30c$ (Eq. 1)
2. $120 = 10a + 30b + 100c$ (Eq. 2)
3. $438 = 30a + 100b + 354c$ (Eq. 3)

3. Solve the System

- Divide (Eq. 1) by 5: $7 = a + 2b + 6c \implies a = 7 - 2b - 6c$ (Eq. 4)
- Substitute (Eq. 4) into (Eq. 2):

$$120 = 10(7 - 2b - 6c) + 30b + 100c$$

$$120 = 70 - 20b - 60c + 30b + 100c$$

$$50 = 10b + 40c \implies 5 = b + 4c \implies b = 5 - 4c \quad (\text{Eq. 5})$$

- Substitute (Eq. 4) into (Eq. 3):

$$438 = 30(7 - 2b - 6c) + 100b + 354c$$

$$438 = 210 - 60b - 180c + 100b + 354c$$

$$228 = 40b + 174c \quad (\text{Eq. 6})$$

- Substitute (Eq. 5) into (Eq. 6):

$$228 = 40(5 - 4c) + 174c$$

$$228 = 200 - 160c + 174c$$

$$28 = 14c \implies c = 2$$

- Find b using $c = 2$ in (Eq. 5):

$$b = 5 - 4(2) = 5 - 8 \implies b = -3$$

- Find a using $b = -3$ and $c = 2$ in (Eq. 4):

$$a = 7 - 2(-3) - 6(2) = 7 + 6 - 12 \implies a = 1$$

Final Result

The best-fit second degree parabola is:

$$\hat{y} = 1 - 3x + 2x^2$$

or

Q2. Find the best values of a & b for $y = ae^{bx}$ by the method of least squares to the following data:

$$\text{Model: } \hat{y} = ae^{bx}$$

X	0	5	8	12	20
y	3.0	1.5	1.0	0.55	0.18

1. Linearization

Take the natural logarithm ($\ln$) of both sides:

$$\ln(\hat{y}) = \ln(ae^{bx})$$

$$\ln(\hat{y}) = \ln(a) + bx$$

Define the new variables and parameters for the linear form $Y = A + BX$:

$$Y = \ln(y), \quad A = \ln(a), \quad B = b, \quad X = x$$

2. Calculate the Required Sums

We need sums of X , Y , X^2 , and XY . $n = 5$.

x (X)	y	$Y = \ln(y)$	X^2	$XY = x \cdot \ln(y)$	
0	3.0	1.0986	0	0.0000	
5	1.5	0.4055	25	2.0275	
8	1.0	0.0000	64	0.0000	

$x \ (X)$	y	$Y = \ln(y)$	X^2	$XY = x \cdot \ln(y)$	
12	0.55	-0.5978	144	-7.1736	
20	0.18	-1.7148	400	-34.2960	
Total		$\sum X = 45$	$\sum Y = -0.8085$	$\sum X^2 = 633$	$\sum XY = -39.4421$

3. Set Up and Solve Normal Equations

- $\sum Y = nA + B \sum X \implies -0.8085 = 5A + 45B \quad (\text{Eq. 1})$
- $\sum XY = A \sum X + B \sum X^2 \implies -39.4421 = 45A + 633B \quad (\text{Eq. 2})$

- Multiply (Eq. 1) by 9 to match the A coefficient in (Eq. 2):

$$\begin{aligned}
 9 \times (-0.8085) &= 9 \times (5A) + 9 \times (45B) \\
 -7.2765 &= 45A + 405B \quad (\text{Eq. 3})
 \end{aligned}$$

- Subtract (Eq. 3) from (Eq. 2):

$$\begin{aligned}
 (-39.4421 - (-7.2765)) &= (45A - 45A) + (633B - 405B) \\
 -32.1656 &= 228B \\
 \mathbf{B} &= \mathbf{-0.14108}
 \end{aligned}$$

- Substitute B back into (Eq. 1):

$$\begin{aligned}
 -0.8085 &= 5A + 45(-0.14108) \\
 -0.8085 &= 5A - 6.3486 \\
 5A &= 6.3486 - 0.8085 = 5.5401 \\
 \mathbf{A} &= \mathbf{1.10802}
 \end{aligned}$$

4. Find Original Parameters (a and b)

- $\mathbf{b} = B \approx -0.1411$
- $\mathbf{a} = e^A = e^{1.10802} \approx \mathbf{3.0286}$

Final Result

The best-fit parameters are:

$$a \approx 3.0286 \quad \text{and} \quad b \approx -0.1411$$

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Weekly Assignment (Curve Fitting)

Q1. & Q4. Fit a straight line trend by the method of least square to the following data:

These two questions are structurally identical to the first example, but use **Year** as the independent variable. To simplify the calculation, we use a technique called **Coding**.

The model is $\hat{y} = a + bx$. The Normal Equations remain the same.

Q1. Trend Fitting (Odd Number of Years)

Year	1979	1980	1981	1982	1983	1984
Production (y)	5	7	9	10	12	17

- $n = 6$ (Even).
- **Coding:** Let X be the coded time variable. Since n is even, the origin must be between the two middle years (1981 and 1982).
- We use the transformation $X = \text{Year} - 1981.5$. This ensures $\sum X = 0$.

Year	X (Year - 1981.5)	y	X^2	XY
1979	-2.5	5	6.25	-12.5
1980	-1.5	7	2.25	-10.5
1981	-0.5	9	0.25	-4.5
1982	0.5	10	0.25	5.0
1983	1.5	12	2.25	18.0
1984	2.5	17	6.25	42.5

Year	X (Year - 1981.5)	y	X^2	XY
Total	$\sum X = 0$	$\sum y = 60$	$\sum X^2 = 17.5$	$\sum XY = 38.0$

- **Normal Equations with $\sum X = 0$:**

$$1. \sum y = na \implies 60 = 6a \implies \mathbf{a = 10}$$

$$2. \sum XY = b \sum X^2 \implies 38.0 = 17.5b \implies \mathbf{b \approx 2.1714}$$

- **Straight Line Trend:**

$$\hat{y} = 10 + 2.1714X$$

(Where $X = \text{Year} - 1981.5$)

Q4. Trend Fitting (Odd Number of Years)

Year	2012	2013	2014	2015	2016	2017
Sales (y)	7	10	12	14	17	24

- $n = 6$ (Even).
- **Coding:** Let $X = \text{Year} - 2014.5$. This also ensures $\sum X = 0$.

Year	X (Year - 2014.5)	y	X^2	XY
2012	-2.5	7	6.25	-17.5
2013	-1.5	10	2.25	-15.0
2014	-0.5	12	0.25	-6.0
2015	0.5	14	0.25	7.0
2016	1.5	17	2.25	25.5
2017	2.5	24	6.25	60.0
Total	$\sum X = 0$	$\sum y = 84$	$\sum X^2 = 17.5$	$\sum XY = 54.0$

- **Normal Equations with $\sum X = 0$:**

$$1. \sum y = na \implies 84 = 6a \implies \mathbf{a = 14}$$

$$2. \sum XY = b \sum X^2 \implies 54.0 = 17.5b \implies \mathbf{b \approx 3.0857}$$

- **Straight Line Trend:**

$$\hat{y} = 14 + 3.0857X$$

(Where $X = \text{Year} - 2014.5$)

»

Expected Questions for University Exam (Curve Fitting)

Q1. Obtain normal equation by method of least square to the curve $y = c_0x + \frac{c_1}{\sqrt{x}}$. Fit it to the following data:

$$\text{Model: } \hat{y} = C_0x + C_1 \frac{1}{\sqrt{x}}$$

X	0.1	0.2	0.4	0.5	1	2
y	21	11	7	6	5	6

1. Obtain Normal Equations

This model is linear in parameters C_0 and C_1 .

- **Coefficient of C_0 :** x
- **Coefficient of C_1 :** $\frac{1}{\sqrt{x}}$

1. **Multiply by x (coefficient of C_0):**

$$\sum (y \cdot x) = C_0 \sum (x \cdot x) + C_1 \sum \left(\frac{1}{\sqrt{x}} \cdot x \right)$$

$$\sum xy = C_0 \sum x^2 + C_1 \sum \sqrt{x}$$

2. **Multiply by $\frac{1}{\sqrt{x}}$ (coefficient of C_1):**

$$\sum \left(y \cdot \frac{1}{\sqrt{x}} \right) = C_0 \sum \left(x \cdot \frac{1}{\sqrt{x}} \right) + C_1 \sum \left(\frac{1}{\sqrt{x}} \cdot \frac{1}{\sqrt{x}} \right)$$

$$\sum \frac{y}{\sqrt{x}} = C_0 \sum \sqrt{x} + C_1 \sum \frac{1}{x}$$

2. Calculate the Required Sums

We need sums of x^2 , $\sqrt{x}$, $\frac{1}{x}$, xy , and $\frac{y}{\sqrt{x}}$. $n = 6$.

x	y	$\sqrt{x}$	x^2	$\frac{1}{x}$	xy	$\frac{y}{\sqrt{x}}$
0.1	21	0.3162	0.01	10.000	2.10	66.398
0.2	11	0.4472	0.04	5.000	2.20	24.597
0.4	7	0.6325	0.16	2.500	2.80	11.067
0.5	6	0.7071	0.25	2.000	3.00	8.485
1.0	5	1.0000	1.00	1.000	5.00	5.000
2.0	6	1.4142	4.00	0.500	12.00	4.243
Total		$\sum x = 4.2$	$\sum \sqrt{x} = 4.5130$	$\sum x^2 = 5.46$	$\sum \frac{1}{x} = 21.000$	$\sum xy = 27.10$

3. Set Up and Solve Normal Equations

$$1. \sum xy = C_0 \sum x^2 + C_1 \sum \sqrt{x}$$

$$27.10 = 5.46C_0 + 4.513C_1 \quad (\text{Eq. 1})$$

$$2. \sum \frac{y}{\sqrt{x}} = C_0 \sum \sqrt{x} + C_1 \sum \frac{1}{x}$$

$$119.79 = 4.513C_0 + 21.00C_1 \quad (\text{Eq. 2})$$

Solving the system (using a precise calculator):

$$C_0 \approx 5.513$$

$$C_1 \approx 4.542$$

Final Result

The best-fit curve is:

$$\hat{y} = 5.513x + 4.542\frac{1}{\sqrt{x}}$$